

### **1. What is Data Science? Explain its primary objective.**

**Answer:**

Data Science is the process of collecting, analyzing, and interpreting data to gain useful insights.

**Primary Objective:** To help make better decisions using data.

### **2. Why is Data Science important in today's digital world? Mention any four reasons.**

**Answer:**

- Helps in better decision-making.
- Predicts future trends.
- Improves business performance.
- Automates tasks using AI.

### **3. List any four benefits of Data Science in business and society.**

**Answer:**

- Better customer service.
- Increased business profits.
- Improved healthcare.
- Fraud detection.

### **4. Mention four real-world applications of Data Science.**

**Answer:**

- Healthcare
- Banking
- E-commerce
- Transportation

### **5. What are the different facets of data? Briefly explain each.**

**Answer:**

- **Structured Data:** Organized in tables.
- **Semi-Structured Data:** Partly organized (XML, JSON).
- **Unstructured Data:** No fixed format (images, videos).
- **Metadata:** Data about data.

**6. Differentiate between structured, semi-structured, and unstructured data with suitable examples.**

**Answer:**

| Type            | Example                |
|-----------------|------------------------|
| Structured      | Student database       |
| Semi-Structured | XML, JSON              |
| Unstructured    | Images, videos, emails |

**7. What is Big Data? State its key characteristics (5 Vs).**

**Answer:**

Big Data is very large and complex data that cannot be processed by traditional tools.

**5 Vs:**

- Volume
- Velocity
- Variety
- Veracity
- Value

**8. Explain the Big Data ecosystem. Name any four technologies used in it.**

**Answer:**

The Big Data ecosystem includes tools for storing, processing, and analyzing large data.

**Technologies:**

- Hadoop
- Spark
- Hive
- HBase

**9. What are the major phases of the Data Science process?**

**Answer:**

1. Data Collection

2. Data Cleaning
3. Data Analysis
4. Model Building
5. Model Evaluation

**10. What is data retrieval? Mention two common sources from which data can be retrieved.**

**Answer:**

Data retrieval means collecting data from different sources.

**Sources:**

- Databases
- Websites/APIs

**11. Why is data collection considered an important step in the Data Science process?**

**Answer:**

Because accurate and relevant data leads to better analysis and results.

**12. What is data cleansing? List any four common data-cleaning techniques.**

**Answer:**

Data cleansing is the process of removing errors and improving data quality.

**Techniques:**

- Remove duplicates
- Handle missing values
- Correct errors
- Remove outliers

**13. What is data integration? Why is it necessary in Data Science?**

**Answer:**

Data integration combines data from different sources into one dataset.

**Need:** It improves accuracy and provides complete information.

**14. Explain data transformation with suitable examples.**

**Answer:**

Data transformation changes data into a useful format.

**Example:** Converting text into numbers or changing date formats.

**15. What is data analysis? Mention any four techniques used for data analysis.**

**Answer:**

Data analysis is the process of examining data to find useful information.

**Techniques:**

- Classification
- Regression
- Clustering
- Visualization

**16. Differentiate between descriptive, predictive, and prescriptive analytics.**

**Answer:**

| Type         | Purpose              |
|--------------|----------------------|
| Descriptive  | What happened?       |
| Predictive   | What may happen?     |
| Prescriptive | What should be done? |

**17. What is meant by building a model in Data Science? Mention the basic steps involved.**

**Answer:**

Model building means creating a machine learning model to make predictions.

**Steps:**

- Select algorithm
- Train model
- Test model
- Improve model

**18. Why is model evaluation important? Name any four evaluation metrics used for classification models.**

**Answer:**

Model evaluation checks how well the model performs.

**Metrics:**

- Accuracy
- Precision
- Recall
- F1-Score

**19. Why is presenting findings an essential part of the Data Science process? Mention any four visualization tools or techniques.****Answer:**

It helps people understand the results and make decisions.

**Tools/Techniques:**

- Bar Chart
- Pie Chart
- Line Chart
- Tableau

**20. How are Data Science models deployed into real-world applications? Give any four examples of Data Science applications.****Answer:**

Models are deployed through web applications, mobile apps, or cloud services.

**Applications:**

- Recommendation systems
- Fraud detection
- Disease prediction
- Weather forecasting